

Excluded Study ID	Primary Reason for Exclusion
E1	Outside the predefined scope of the review.
E2	Not aligned with the technical scope defined by the research questions.
E3	Did not address the technical components covered by the review.
E4	Outside the targeted scope of IoT-based smart waste management.
E5	Did not meet the predefined technical-scope eligibility criterion.
E6	Focused on a waste-management topic outside the scope of the synthesis.
E7	Did not address the smart waste-management components targeted in the review.
E8	Outside the predefined scope for sensing, forecasting, routing optimization, or digital-twin applications.
E9	Did not provide evidence relevant to the technical scope of the review.
E10	Not sufficiently aligned with the predefined smart waste-management scope.
E11	Addressed a topic outside the technical focus established for the review.
E12	Did not fall within the technical areas covered by the five research questions.
E13	Outside the predefined application scope of the review.
E14	Did not address the targeted IoT, AI, optimization, or digital-twin components.
E15	Not relevant to the predefined technical focus of the synthesis.
E16	Addressed waste management but fell outside the scope specified by the eligibility criteria.
E17	Did not satisfy the predefined scope requirements for inclusion.
E18	Outside the targeted technical context of smart waste collection and management.
E19	Did not address any of the core technical components considered in the review.
E20	Outside the predefined review scope and therefore not eligible for the final synthesis.
E21	Proposed technical approach without sufficient experimental or simulation-based validation.
E22	Proposed framework without evidence-based evaluation.
E23	Technical approach was not supported by sufficient validation evidence.
E24	Proposed method without experimental, simulation, or case-study validation.
E25	Presented an unvalidated technical approach that did not meet the predefined eligibility criterion.
E26	Reported technical claims without sufficient evidence-based assessment.
E27	Insufficient methodological detail for evidence extraction and comparison.
E28	Technical approach was not described in sufficient detail for inclusion in the synthesis.
E29	Insufficient information was provided to identify the main technical approach and evaluation setup.
E30	Did not provide sufficient methodological information to apply the predefined coding framework.
E31	Did not meet the predefined criterion for sufficient methodological detail.
E32	Review-only article without original implementation evidence.
E33	Review article summarizing previous studies without an original technical implementation.
E34	Literature-based synthesis without original experimental, simulation, or deployment evidence.
E35	Review-only publication without original implementation or technical evaluation evidence.